

Bureau of Indian Standards

Manak Bhawan, 9, Bahadur Shah Zafar Marg
New Delhi - 110002

IS 1489:1991

1991

PORTLAND POZZOLANA CEMENT

SPECIFICATION

ICS 91.100.10

BIS 1991

Indian Standard

1 SCOPE

This standard lays down the requirements for Portland Pozzolana Cement (PPC). Portland Pozzolana Cement is manufactured by inter-grinding Portland cement clinker with pozzolanic materials such as fly ash, calcined clay, or volcanic ash, together with gypsum. This standard covers three grades: 33, 43, and 53.

2 REFERENCES

- IS 269:2015 - Ordinary Portland Cement, 33 Grade - Specification
- IS 383:2016 - Coarse and Fine Aggregates for Concrete
- IS 4031 (Part 1 to 15) - Methods of Physical and Chemical Tests for Cement
- IS 1489 (Part 1):1991 - Portland Pozzolana Cement, Part 1: Fly Ash Based
- IS 1489 (Part 2):1991 - Portland Pozzolana Cement, Part 2: Calcined Clay Based

3 DEFINITIONS

3.1 Portland Pozzolana Cement

Portland Pozzolana Cement is a hydraulic cement produced by inter-grinding Portland cement clinker with pozzolanic materials and gypsum. The pozzolanic material reacts with calcium hydroxide liberated during cement hydration to form additional calcium silicate hydrate, improving long-term strength and durability.

4 REQUIREMENTS

The cement shall conform to the requirements given in Table 1.

S.No.	Characteristic	33 Grade	43 Grade	53 Grade
1	Fineness (Blaine), m/kg, min.	300	300	300
2	Soundness (Le Chatelier), mm, max.	10	10	10
3	Setting Time (Vicat), min.			
	Initial, min.	30	30	30
	Final, max., min.	600	600	600
4	Compressive Strength, MPa			
	3 days, min.	10	23	27
	7 days, min.	16	33	37
	28 days, min.	33	43	53
5	Autoclave expansion, %, max.	0.8	0.8	0.8

5 COMPOSITION

5.1 Pozzolana Content

The percentage of pozzolana (fly ash or calcined clay) shall be between 15% and 35% by mass of Portland Pozzolana Cement. For fly ash based PPC, the fly ash shall conform to IS 3812 (Part 1).

5.2 Gypsum

The amount of gypsum added during grinding shall not exceed 5% by mass of the cement, calculated as calcium sulphate dihydrate ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$).

6 TEST METHODS

The methods of test for determining the various characteristics of cement shall be in accordance with IS 4031 (Part 1 to 15).

- Fineness: IS 4031 (Part 1) - Dry Sieve Method or Blaine Air Permeability
- Soundness: IS 4031 (Part 3) - Le Chatelier Method
- Setting Time: IS 4031 (Part 5) - Vicat Apparatus
- Compressive Strength: IS 4031 (Part 6) - Cube Moulds (70.6 mm)
- Autoclave Expansion: IS 4031 (Part 9)

7 MARKING

Each bag or container of cement shall be clearly marked with the following information: (a) Name and trade mark of the manufacturer, (b) Name of the standard: IS 1489, (c) Grade of cement, (d) Net weight of cement, (e) Batch number, (f) Month and year of manufacture.

8 PACKING AND STORAGE

Cement shall be packed in bags of 50 kg nominal mass. The cement shall be stored in dry conditions, protected from moisture. Storage silos or bins shall have proper ventilation to prevent caking.